

C-Mem MBR (Membrane Bioreactor)

Advanced Wastewater Recycling with Membrane Technology.

C-MEM Operating Cycle

Cycle parameter	Preliminary value
Filtration period	9 minutes
Relaxation period	1 minute
Total cycle	10 minutes
Filtration fraction	90%
Relaxation fraction	10%
Air scour	Continuous or intermittent during operation
Backwash	Automatic, based on TMP or operating time
Maintenance cleaning	Approximately weekly, subject to fouling rate
CIP	As required based on permeability recovery

Membrane Technical Data

Parameter	C-MEM MBR data
Membrane process	Submerged ultrafiltration
Membrane configuration	Hollow fibre
Membrane material	Organic hydrophilic HDPE
Current product description	PFAS-free HDPE hollow fibre
Fibre outside diameter	Approximately 0.3-0.5 mm
Average pore size	Approximately 20.1 nm
Equivalent pore size	Approximately 0.02 µm
Filtration direction	Outside-in
Normal application	Submerged membrane bioreactor
Submerged TMP	Approximately 50-600 mbar
Pressure application TMP	Up to approximately 3 bar
Clean-water flux	Up to approximately 300 L/m ² ·h
Wastewater design flux	Must be confirmed by process design and pilot testing
Membrane character	Permanently hydrophilic and anti-fouling
Disinfection barrier	Physical barrier for suspended solids, bacteria and some viruses
Membrane support	Fibres wound around a carrier cartridge
Membrane operation	Suction filtration with air scour
Permeate withdrawal	Through the internal lumen of the hollow fibres